



Mizan — Shared Credit Decisioning Layer · PRD & Execution Roadmap

Mizan (ميزان — "the scales") is the working codename for Mal Bank's product-agnostic credit decisioning service: one platform that any lending product — retail or SME — calls to get an eligibility decision, a facility limit, and pricing guidance.

Field	Value
Document owner	Evgeny Muravev
Status	Draft v1.0 — for panel review
Product stage	0 → 1 (internal platform product)
Date	July 2026
Anchor journeys	Retail personal finance (Murabaha/Tawarruq) · SME working capital (revolving Tawarruq)
Consumers of this service	Retail lending squad, SME lending squad, future product teams (auto finance, trade finance)
Key stakeholders	CRO / Credit Risk, ISSC & Shari'ah Compliance, Compliance & Consumer Protection, Data/Analytics, Core Banking, Operations (credit review)

How to read this in 20 minutes: §1 Executive Summary → §5 Functional Architecture → §7 Trade-offs → §9 Roadmap. Everything else (§8 Requirements, §10 Backlog, §11 Integrations) is the supporting depth.

1. Executive Summary

Mal Bank runs two lending portfolios on largely bespoke scoring and underwriting logic. As both scale, every new product re-implements the same plumbing — bureau consent, AECB pulls, affordability computation, decision audit — while the genuinely different parts (retail vs SME underwriting judgment) get entangled with that plumbing and become slow to change.

Mizan is a shared decisioning layer with a simple contract: a product journey submits an application; Mizan orchestrates consent, data acquisition (AECB + open-finance/bank data + internal data), affordability, policy rules, and scoring; and returns **approve / refer / decline + facility limit + risk grade + pricing band**, with every decision fully replayable for audit — CBUAE model-risk audit *and* Shari'ah audit.

The six positions this document takes (argued in §7):

- Share the pipes, not the brain.** One decisioning platform and one data layer; segment-specific policy packs and scorecards for retail vs SME. No single unified scoring model.
- Buy the decision engine, own the contract.** License a configurable decisioning platform (deployed in-UAE per the Outsourcing Regulation); build the decision API, adapters, feature definitions, and decision record store in-house so the vendor stays swappable.
- Real-time bureau, cached and reusable.** Real-time AECB API pulls at application with a consent-bound cache; monthly furnishing is a separate compliance pipeline.
- Shariah-aware, not Shariah-executing.** Mizan issues a time-boxed approval; Murabaha/Tawarruq contract execution (wakala → commodity purchase → offer/acceptance) stays in the product layer. Mizan's decision log

links to contract events so the full sequence is auditable.

5. **Consent is a first-class shared service.** A central consent registry (AECB + AI Tareq open finance) is the prerequisite for the highest-value future capability: continuous re-scoring and limit management.
6. **Refer is a designed path, not a failure mode.** SME lending will launch refer-heavy by design; the workbench, SLAs and override governance are core scope, not afterthoughts.

Why now: the UAE data infrastructure has just crossed the threshold that makes this layer worth building. Open Finance went live under AI Tareq in January 2026 (FAB, CBD, ADIB), AECB's Credit Score 3.0 launches in 2026 with alternative-data scores for thin files, e-invoicing becomes mandatory from 2027, and the new SME Customer Protection Regulation (Circular 2/2026) raises the conduct bar on SME lending. A bank that wires these into one decisioning layer now will originate faster and safer than one that retrofits later.

Strategic frame. Mizan is not a neutral utility — it is the engine that operationalizes Mal's credit portfolio strategy (companion document: [Cross-Border Underwriting Dossier](#)). Three strategy commitments shape the platform's design beyond the anchor journeys: **(a) cross-border data is the differentiated asset** — domestic aggregation commoditizes as AI Tareq phases in, so the adapters that matter most are the ones that see the *other* country of an expat's life (home-country bureau via the AECB × Nova Credit Passport rail, home-country accounts, remittance behavior); **(b) intent is underwriting material** — the agentic PFM platform generates goal-pursuit telemetry (a shadow instalment history and the earliest distress sensor there is), which Mizan ingests as a governed, overlay-only feature stream; **(c) risk is managed at day zero and recourse is structured at origination** — early-warning signals fire same-day, and approvals carry machine-readable, consent-based recourse conditions (salary/EOSB assignment, takaful, remittance-linked repayment) rather than improvised collections pressure later.

2. Context, Problem, Goals

Context

Mal Bank operates retail consumer finance (personal finance/Murabaha, salary advance/Qard Hassan, auto finance) and SME lending (working capital, term, trade finance) as parallel stacks. Each product today embeds its own scoring calls, policy logic and manual review flow. This made sense at launch; at scale it produces three compounding costs: duplicated integration spend, inconsistent risk treatment of the same customer across products, and — most damaging — a credit policy iteration cycle gated on engineering releases.

Meanwhile the market context is moving: digital-native competitors (Wio in SME, ruya in Islamic retail finance) are originating with instant or near-instant decisions; ruya already executes commodity Murabaha straight-through via DDCAP's ETHOS platform. Regulatory infrastructure (AI Tareq, AECB 3.0, e-invoicing) is turning data that used to be documents into APIs.

The portfolio strategy this layer serves is set out in the companion dossier: the UAE's ~60 licensed banks all underwrite the onshore half of a predominantly expatriate customer base, pricing the "skip" blind as a margin tax. Mal's edge is consented cross-border visibility — income, obligations, credit history and *intentions* in both countries of the borrower's life — and a product sequence ordered by learning speed (fast-seasoning, granular, Shari'ah-native products first; mortgage refinance last, as retention). Mizan is where that strategy becomes executable: the cross-border adapters, the intent-feature governance, the day-zero early-warning loop and origination-time recourse conditions all live in this layer.

Problem

Every new lending decision at Mal Bank re-builds the same underwriting plumbing — slowly, inconsistently, and in a way Credit Risk cannot tune without an engineering release.

Goals

1. **One decision API** any lending product calls – retail personal finance and SME working capital live on it within the roadmap horizon; a third product onboards in weeks, not months.
2. **Credit Risk self-service**: policy parameters, cut-offs and challenger strategies changed by the risk team in days, under governance, without code deployment.
3. **Automation with control**: $\geq 80\%$ straight-through processing (STP) on retail personal finance by month 9; SME working capital launched refer-first, reaching $\geq 40\%$ STP by month 12.
4. **Every decision replayable**: 100% of decisions reconstructable – inputs, features, rules fired, model version, human overrides – satisfying CBUAE Model Management Standards, the 2026 AI/ML guidance note, and internal/external Shari'ah audit.

Non-goals (explicitly out of scope)

- Origination UX, KYC/onboarding, and contract execution (Murabaha/Tawarruq mechanics) – Mizan integrates with these, it does not own them.
- Fraud decisioning and AML screening – separate services; Mizan consumes their outcomes as inputs.
- Collections strategy and recovery scoring – adjacent roadmap, not in this document (interface points noted).
- Deposit/investment product decisioning.

Definition of Done (for this roadmap)

Both anchor journeys decision through Mizan in production; Credit Risk has shipped ≥ 3 policy changes without engineering; a third product's decisioning has been onboarded using only configuration and adapters – no changes to the core.

Value proposition

Customers (retail & SME): decisions in minutes not days; one consent experience; consistent treatment across products; transparent decline reasons (AECB 3.0-aligned reason codes, Arabic + English).

Mal Bank: faster product launches on shared rails; lower cost per decision; a single governed control point for CBUAE model-risk, consumer-protection and Shari'ah obligations; portfolio-level risk visibility across segments.

Credit Risk team: ownership of policy without engineering dependency; champion/challenger experimentation; one decision dataset to build future scorecards on.

3. Glossary

Term	Definition
Murabaha	Cost-plus sale (not a loan): the bank acquires an asset and sells it to the customer at disclosed cost + profit, paid deferred. Governed by AAOIFI Shari'ah Standard 8; codified in UAE Commercial Transactions Law (Arts. 468–497).
Tawarruq / commodity Murabaha	Monetization structure behind "cash" finance: bank buys a commodity, sells it to the customer on deferred Murabaha terms, customer (usually via agency) sells it for spot cash. Governed by AAOIFI SS 30.
Qard Hassan	Benevolent interest-free loan (basis of salary advance). Only a flat admin fee equal to actual service cost is permitted – the fee may not scale with amount or tenor (AAOIFI SS 19).
Wakala	Agency contract; in Tawarruq, the customer typically appoints the bank as agent to sell the commodity. Must be a contract separate from the Murabaha.

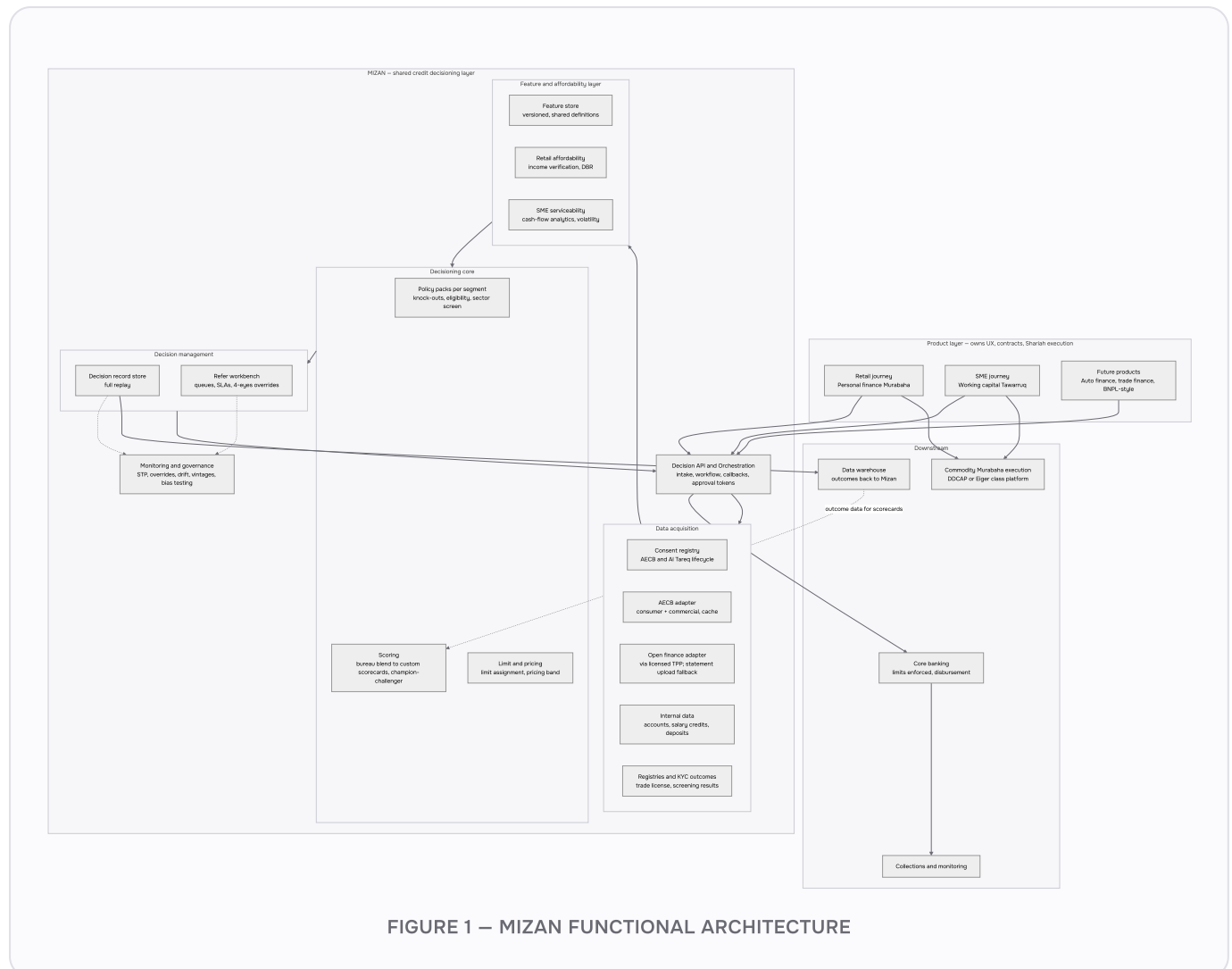
Term	Definition
AECB	AI Etihad Credit Bureau — UAE federal credit bureau. Consumer & commercial reports and scores (300–900). Consent-based access under Federal Law 6/2010; mandatory pull before granting credit; mandatory monthly furnishing.
AI Tareq / Nebras	UAE Open Finance framework: Nebras Open Finance LLC (CBUAE / AI Etihad Payments subsidiary) operates the central API hub and trust framework; "AI Tareq" is the consumer-facing consent brand. Regulation: Circular 3/2025 (replacing 7/2023).
OFP / TPP	Open Finance Provider — the CBUAE license category for third parties consuming open-finance data / initiating services. A bank may itself obtain OFP authorization (ADIB, April 2026).
DBR	Debt Burden Ratio — total monthly repayment obligations ÷ gross salary + regular income. Regulatory cap 50% (30% for retirees).
STP	Straight-through processing — decisions completed with no human touch.
Refer	Decision outcome routing an application to a human credit analyst with a structured worklist and SLA.
HSA / ISSC	Higher Shari'ah Authority (CBUAE, rulings binding on all Islamic financial institutions) / Internal Shari'ah Supervision Committee (Mal Bank's own board-level Shari'ah committee; its fatwas bind the bank). AAOIFI standards are mandatory minimums in the UAE since 1 Sept 2018.
MMS	CBUAE Model Management Standards (Dec 2022) — mandatory lifecycle governance for all models, including credit scoring: Model Oversight Committee, independent validation, prescribed monitoring.
KFS	Key Facts Statement — mandatory plain-language product disclosure (Arabic + English) under the Consumer Protection Standards.
ESR	AECB's Expense-to-Salary Ratio indicator — bureau-computed obligations vs reported salary, included in credit reports since 2022.

4. Guiding Principles

- 1. Product-agnostic core, product-specific configuration.** A new product = a *product manifest* (data requirements, policy pack, scorecard binding, limit logic) — never a change to the core engine.
- 2. Decisions are data.** The decision record — not the loan account — is the atomic asset: versioned inputs, features, rules, models, outcomes. It feeds audit, scorecard development, and regulatory reporting.
- 3. Regulatory rules are code-reviewed policy primitives.** $DBR \leq 50\%$, 20× salary cap, 48-month tenor, mandatory AECB check, sector exclusions — encoded once, centrally, tested, and reused by every policy pack. Products cannot opt out.
- 4. Human judgment is part of the system.** Overrides are structured (reason-coded, 4-eyes, monitored), and override analytics feed policy iteration.
- 5. Degrade gracefully.** Every external dependency (AECB, AI Tareq, registries) has a defined fallback: cached data, document upload, or refer — never a hard journey failure.
- 6. Bilingual and explainable by design.** Every decline/refer carries customer-safe reason codes renderable in Arabic and English — a 2026 AI/ML guidance-note requirement, and the right thing to do.

5. Functional Architecture

5.1 The picture



5.2 What is shared vs segment-specific

Capability	Shared	Retail-specific	SME-specific
Decision API contract, orchestration, approval tokens	✓	—	—
Consent registry (AECB + open finance)	✓	—	—
AECB adapter	✓	Consumer report + score, ESR, Credit Passport for thin files	Commercial report + score plus personal reports of shareholders/guarantors (blended view)
Open finance / bank data ingestion	✓	Salary account transactions	Business account cash-flow, multi- bank

Capability	Shared	Retail-specific	SME-specific
Feature store & categorization	✓ engine	Salary detection, obligations	Inflow stability, concentration, seasonality, bounced-cheque signals
Affordability	✓ framework	DBR $\leq$ 50% (30% retiree), 20× salary, 48-month tenor	Serviceability from verified net cash-flow; owner obligations counted
Policy packs	✓ engine + regulatory primitives	Retail pack (age, residency, salary floor, employer risk)	SME pack (license validity/age, sector screen incl. Shari'ah exclusions, entity age, cheque history)
Scorecards	✓ hosting, registry, champion/challenger	Retail application scorecard (bureau-anchored v0)	SME scorecard (owner score + commercial score + cash-flow features)
Limit & pricing	✓ framework	Limit = min(DBR headroom, 20× salary, product cap); pricing band	Limit = f(verified monthly net inflows, tenor); facility + drawdown structure
Refer workbench, decision store, monitoring	✓	Queue config	Queue config, RM attachments
Cross-border data adapters	✓ engine	Credit Passport (home bureau via AECB × Nova), home-country accounts/statements, remittance behavior	Cross-border supplier/buyer flows, trade counterparty data
Goal-pursuit / intent telemetry (from PFM agent)	✓ ingestion + governance	Goal-contribution consistency, goal-behavior delta, distress signal (contributions stop)	Plan-vs-actual variance, use-of-proceeds tracking, milestone hit-rate
Recourse-conditions engine	✓ framework	Salary + EOSB assignment, takaful, remittance-linked repayment	Kafalah/guarantor structuring, personal guarantees

The sharing prize is the left column — roughly 70% of build effort — not a unified model. Retail and SME underwriting genuinely differ; forcing one scoring engine across both would be worse at each and slower to iterate.

5.3 Anchor journey 1 — Retail personal finance (Murabaha/Tawarruq)

1. Customer applies in-app; product layer captures amount/tenor and **consent** (AECB pull consent per Federal Law 6/2010 + open-finance consent where used) → registered in the consent registry.
2. Mizan orchestration validates KYC/screening outcomes (consumed, not performed), pulls **AECB consumer report + score** in real time (cache TTL applied), acquires salary/transaction data (internal salary-transfer history where Mal Bank is the salary bank; Al Tareq via TPP otherwise; ESR as cross-check).
3. Feature layer verifies income, computes obligations (including guarantees) and **DBR**. Thin-file applicants route to the **cross-border path, not an exception queue**: with consent, the Credit Passport pull (home-country bureau via AECB × Nova Credit) converts a no-hit into an underwritable file — per the strategy dossier, this population is a target approval segment (Amex's Credit Passport-approved newcomers ran 79% less risky than domestic prime), approved conservatively with recourse conditions rather than referred by default.
4. Retail policy pack runs knock-outs (regulatory primitives + bank policy); scorecard assigns risk grade; limit engine computes max facility = min(DBR headroom at proposed tenor, 20× salary, product cap); pricing band selected by grade (banded, not individual — see §7.5).

5. Mizan returns **approve + limit + pricing band + approval token** (validity window, conditions) in < 5 minutes p90, or refer/decline with reason codes. The token's conditions include **origination-time recourse**: salary-transfer + EOSB assignment, credit-shield takaful enrollment, and — for cross-border-underwritten approvals — a remittance-linked repayment schedule (instalment settles before the home transfer leaves). Consent-based recourse structured here is what makes day-zero risk management possible later; it is never improvised at delinquency.
6. **Product layer executes the Shari'ah leg**: customer promise, wakala, commodity purchase and Murabaha offer/acceptance via an execution platform (DDCAP/Eiger-class), then disbursement. *The debt legally exists only after this sequence — Mizan's approval is a risk decision, not a contract.* Contract event IDs are written back to the decision record so Shari'ah audit can replay approval → execution ordering end-to-end.
7. Cooling-off (5 business days, waivable in writing) and KFS disclosure are product-layer obligations; Mizan supplies the numbers (profit rate band, total obligation) that populate the KFS.

Salary advance (Qard Hassan) corollary: because AAOIFI SS 19 forbids fees that scale with amount or tenor, there is no pricing lever — decisioning for salary advance is pure eligibility + limit control (% of verified salary, e.g. market practice 80–90% capped). It should be Mizan's simplest policy pack, and it is a cheap early proof of product-agnosticism.

5.4 Anchor journey 2 — SME working capital (revolving Tawarruq)

1. SME applies via business portal or RM; consents captured for **entity and individual** pulls (owners/guarantors).
2. Data acquisition: AECB **commercial** report/score + **personal** reports of major shareholders and guarantors; trade license validity (registry lookup); business banking data — Mal Bank accounts internally, other banks via Al Tareq TPP where live, otherwise **statement upload with automated parsing** (the designed fallback, not an exception); optionally VAT returns as documents at higher ticket sizes.
3. Feature layer computes cash-flow serviceability: verified monthly net inflows, inflow stability and concentration, bounced-cheque history (bureau + ChequeScore-class signals), owner obligations.
4. SME policy pack: regulatory + Shari'ah **sector screen** (ISSC-approved exclusion list seeded from the AAOIFI SS 21 taxonomy: conventional finance, alcohol, gambling, pork, adult entertainment, weapons; tobacco per ISSC ruling), license age/validity, minimum trading history, purpose-of-finance undertaking.
5. Facility-level decision: **approve / refer / decline + facility limit** (e.g., a multiple of verified monthly net inflows, tenor-adjusted) + pricing band. At launch most non-trivial cases **refer by design** to the workbench with a pre-assembled credit file (all data, features and rule results on one screen — the analyst's time goes to judgment, not data gathering). Target: refer resolution < 24h.
6. **Drawdowns**: each revolving drawdown is a fresh commodity-Murabaha cycle executed by the product layer; Mizan performs a lightweight **drawdown check** (limit availability, arrears status, material-deterioration flag) rather than full re-underwriting — this keeps utilization instant while the Shari'ah execution runs straight-through.
7. Ongoing: facility review triggers (behavioral, time-based) route through the same API — the re-scoring foundation (§8.10).

6. Regulatory & Shari'ah Guardrails (what the engine must encode)

These are hard constraints, encoded once as shared policy primitives (§4.3). Sources were verified against CBUAE Rulebook material and current legal commentary; items to re-verify against primary texts during discovery are flagged in §15.

6.1 Regulatory primitives

Rule	Source	Encoding in Mizan
AECB report must be requested before extending credit; borrower data furnished at least monthly	CBUAE Rulebook credit-report articles; Federal Law 6/2010	Bureau pull is a mandatory orchestration step; furnishing pipeline is a P1 epic
Written customer consent to access credit information	Federal Law 6/2010 (as amended)	Consent registry gate – no consent record, no pull, no decision
DBR $\leq$ 50% of gross salary + regular income; $\leq$ 30% for retirees	Reg 29/2011 / CPS	Shared affordability primitive; retiree flag switches cap
Personal finance $\leq$ 20 $\times$ salary; tenor $\leq$ 48 months	Reg 29/2011	Retail limit-engine ceiling
Auto finance $\leq$ 80% LTV, $\leq$ 60 months (future product)	Reg 29/2011	Pre-built primitive for P4 onboarding
Responsible financing: affordability beyond mechanical DBR; suitability	CPS Art. 7	Income verification standards; refer triggers on affordability edge cases
5-business-day cooling-off (waivable in writing); KFS in Arabic + English	CPR 8/2020 / CPS	Product-layer duty; Mizan supplies KFS figures and records decision basis
Fee caps incl. early settlement $\leq$ 1% or AED 10,000	Reg 29/2011 Appendix 2 (as amended 2019)	Pricing engine validation bounds
SME conduct: segmentation, no bundling/tied selling, disclosure and complaint SLAs	SME Customer Protection Regulation, Circular 2/2026 (effective ~H2 2026)	SME policy pack + decision comms templates; launch-blocking compliance review in P2
Model governance: Model Oversight Committee, independent validation, monitoring, full documentation	Model Management Standards (Notice 5052/2022)	Model registry, validation workflow, monitoring dashboards are core scope – not compliance retrofits
AI/ML: explainability for credit decisions, fairness/bias stress-testing, human oversight, bank fully responsible even when outsourced	CBUAE Guidance Note, 11 Feb 2026	Reason codes on every decision; scheduled bias tests; vendor due-diligence pack
Material outsourcing needs prior CBUAE non-objection; confidential customer data stays in-UAE absent CBUAE + customer consent	Outsourcing Regulation & Standards (2021)	Vendor must deploy in-UAE; non-objection filing is a P0 roadmap gate
Purpose limitation on personal data: service data may not be repurposed beyond its consented use	PDPL (Federal Decree-Law 45/2021); CPS data-protection provisions	PFM/goal telemetry enters credit features only under an explicit, separately consented purpose; it is never available to collections as pressure material (no debt disclosure to family, no contact-graph leverage – the dossier's "never" list is encoded as a platform-level access restriction, not a policy hope)

6.2 Shari'ah constraints with direct decisioning impact

Constraint	Source	Product implication
Murabaha sequence is validity-critical: promise → bank acquires asset → offer/acceptance. Approval ≠ debt.	AAOIFI SS 8, SS 30	Mizan issues a time-boxed approval token ; contract events logged against the decision record; sequencing provable in Shari'ah audit
Sale price immutable once contracted; no increase for late payment or extension; rescheduling only without debt increase	AAOIFI SS 8; UAE courts	Re-pricing levers excluded from collections-adjacent logic; restructuring flows cannot touch price
Late payment amounts are charity donations , not income; late-payment interest void even if agreed (Dubai Cassation 595/2025; Cassation 9/2025 voided "interest"-labelled clauses)	AAOIFI SS 3; UAE case law	Terminology is revenue protection, not cosmetics: "profit rate" not interest, "finance" not loan – enforced in reason codes, KFS feeds, all decision comms
Qard Hassan fee = actual service cost only; cannot scale with amount or tenor	AAOIFI SS 19	No risk- or amount-based fee tiers on salary advance; eligibility/limit control only
No sale of debt at discount	AAOIFI SS 59	No discounted NPL sales for capital relief; collections outsourcing fee-based – affects portfolio strategy assumptions, not the engine
Sector screening for SME financing; financed goods must be halal	AAOIFI SS 21 taxonomy + ISSC policy	Machine-readable, ISSC-approved exclusion list versioned in the policy engine; purpose-of-finance undertakings captured
ISSC approval scope covers policies, product structures, contracts and operational process flows; AAOIFI standards binding minimums (HSA Res. 18/3/2018)	CBUAE Shari'ah Governance Standard	New engine, sector screens, pricing/fee mechanics, late-payment handling all need ISSC sign-off. Recommendation: agree a materiality-based delegation matrix with the ISSC up front – e.g., score cut-off recalibration within an approved framework = internal; anything touching contract mechanics, fees or exclusions = ISSC. Without this, the "policy changes in days" goal dies in committee.
Risk-based pricing: no explicit prohibition found; price must be fixed and disclosed at contract execution; ISSC holds a "fairness" remit; large UAE Islamic banks practice category-uniform rates	Doctrine + market practice	Launch with banded pricing (grade → band); revisit individual risk-based markup with ISSC in P3. Engine outputs a band either way, so the decision is reversible.

7. Key Trade-offs & Recommendations

7.1 One unified scoring engine vs segment-specific engines sharing data plumbing

Recommendation: one decisioning *platform*, two decisioning *brains* (policy packs + scorecards per segment). A unified model flattens genuinely different risk logic: retail is salary-anchored, high-volume, thin application data; SME is cash-flow-anchored, heterogeneous, blends entity and owner risk. But separate *platforms* would duplicate the 70% of effort that is identical (consent, bureau, data, audit, workbench, monitoring). Share infrastructure and feature definitions; never force one model. This also mirrors how the org works – one Credit Risk function, different credit committees/appetites per segment.

7.2 Build vs buy the rules/decisioning engine

Recommendation: buy the engine; build the boundary. License a configurable decisioning platform for rules authoring, champion/challenger and simulation — this is exactly the capability that gives Credit Risk self-service, and building it in-house is 12+ months of undifferentiated engineering. Build and own in-house: the decision API contract, integration adapters, feature definitions, and the decision record store. That boundary keeps the vendor swappable and the data ours.

Vendor constraints from research: the CBUAE Outsourcing Regulation requires prior non-objection for material outsourcing and effectively requires customer data to stay in-UAE — so **in-country deployment is a hard requirement**. FICO (hosted in-region via Network International), Experian PowerCurve (Dubai entity, GCC installed base) and Provenir (expanding MEA) clear this most credibly today; Taktile has the best risk-team ergonomics but no public MEA presence or data-residency story — a diligence risk, not a disqualification. Selection is a PO activity with a 6-week time-box: shortlist 3, decision-scenario bake-off (our two anchor journeys as the test script), reference calls, security/residency review.

7.3 Real-time API bureau integration vs batch

Recommendation: real-time, with a consent-bound cache; batch only for furnishing. Real-time AECB pulls are proven in-market (Policybazaar's API integration; Nova Credit's in-flow Credit Passport). Batch pulls would cap us at next-day decisions and kill the instant-Murabaha experience competitors already ship. Cache reports (TTL ~30 days, subject to AECB terms) to avoid duplicate cost when a customer applies for a second product; the mandatory monthly furnishing pipeline is separate compliance plumbing, built once in P1.

7.4 Open finance access: own OFP/TPP license vs licensed TPP partner vs documents

Recommendation: sequence all three. (a) Launch with a **licensed TPP partner** (Lean is live under Al Tareq; Tarabut has in-principle approval) for account/transaction data plus their categorization and income-verification tooling. (b) Keep **statement upload + parsing** as a designed SME fallback — Al Tareq bank coverage is still ramping in 2026 and SME data needs must not be hostage to it. (c) Evaluate **our own OFP authorization** in P3 once volumes justify it — ADIB proved the path in April 2026; owning the license removes per-call TPP economics and enables the re-scoring use case at portfolio scale.

7.5 Risk-based pricing vs banded pricing

Recommendation: banded at launch, individual later (maybe). Nothing in AAOIFI prohibits differentiated Murabaha markups, but conservative market practice (ADIB's category-uniform rates) and the ISSC's fairness remit argue for launching with grade-based bands. The engine outputs `pricing_band` from day one, so moving to finer-grained pricing is a policy decision with ISSC, not a rebuild.

7.6 Where the limit ledger lives

Recommendation: Mizan decides, core banking enforces. Mizan owns limit *derivation* and the decision history behind it; the core banking system remains the system of record for the booked facility and enforces utilization. Duplicating enforcement in Mizan would create reconciliation hell; putting derivation in the core would bury policy in the least agile system we own.

7.7 Intent features: in the scorecard vs overlays on it

Recommendation: overlays on an explainable core, never the core. Goal-pursuit and behavioral features are the strategy's most differentiated signal — and its most governance-sensitive. Three constraints decide the architecture: CBUAE explainability expectations for high-impact credit decisions (2026 AI/ML guidance note), Goodhart risk (once pursuit is a known input it gets simulated — so weight only costly-to-fake flows like actual transfers, and never disclose feature weights), and the never-punish-honesty rule (score inconsistency-without-cause, not goal changes; a declared departure earns a tenor-matched product, not a price penalty — otherwise disclosure stops and the data poisons itself). Intent features therefore enter as bounded adjustments on the

bureau-anchored scorecard, with their marginal contribution monitored and capped. User-controlled imports (e.g., AI-chat profile exports) are onboarding personalization only — unverifiable and adverse-selecting as credit inputs.

8. Product Requirements

Requirement sections are numbered; each has a goal statement and representative stories with acceptance criteria. This is a skeleton — deliberately 2–3 stories per section, not exhaustive.

8.1 Decision API & Orchestration — [P1]

Goal: any product journey can request a credit decision through one stable, versioned contract.

User story	Acceptance criteria
As a product squad, I submit an application and receive a decision or a tracked async outcome	- POST /decisions accepts a product manifest ID + applicant payload; sync response < 5 min p90 for retail - Async callbacks/webhooks for refer outcomes; idempotency keys on all submissions - Decision response = outcome (approve/refer/decline), limit, tenor, risk grade, pricing band, reason codes, approval token with expiry & conditions
As the product layer, I confirm contract execution against an approval	- Execution events (wakala signed, commodity purchased, offer accepted) POST-able against the decision ID - Token expiry configurable per product (e.g., aligned to Murabaha offer-validity windows); expired tokens require re-decision, re-using cached data where consent allows

8.2 Consent & Data Rights — [P1]

Goal: no data is pulled without a valid, evidenced, product-scoped consent — and consent is reusable where lawful.

User story	Acceptance criteria
As a customer, I grant AECEB and open-finance consent digitally and can see/revoke it	- Consent records: scope, purpose, timestamp, evidence (what the customer saw), expiry - AECEB consent captured per Federal Law 6/2010; AI Tareq consent via the central consent/auth layer - Revocation stops future pulls within the same day and is logged
As Mizan, I enforce consent before any external pull	- Orchestration refuses bureau/open-finance calls without a matching active consent; attempted violations alert

8.3 AECEB Integration — [P1 consumer, P2 commercial]

Goal: real-time bureau data for decisions; compliant furnishing back.

User story	Acceptance criteria
As the decision flow, I pull consumer report + score in real time	- p95 pull latency within journey budget; cache with TTL honoring AECEB terms - No-hit/thin-file classified distinctly from adverse data and routed to thin-file strategy (Credit Passport where consented; refer; starter-product cross-offer) - ESR captured as a cross-check feature
As the SME flow, I pull a blended entity + individuals view	- Commercial report/score + personal reports for shareholders/guarantors above ownership threshold, each under its own consent - Blended obligations feed SME serviceability

User story	Acceptance criteria
As a thin-file applicant, my home-country history counts	- Credit Passport pull (AECB × Nova Credit; India/Philippines/UK live) under its own explicit consent - Home-bureau score maps into the scorecard with a conservative overlay and grade cap; approval carries cross-border recourse conditions - Declining Passport consent routes to refer, never auto-decline
As Compliance, portfolio data is furnished monthly	- Automated monthly submission with quality checks and rejection reprocessing; reconciliation report

8.4 Bank Data & Affordability — *[P1 retail basics, P2 SME cash-flow]*

Goal: verified income and obligations for retail; verified cash-flow serviceability for SME.

User story	Acceptance criteria
As the retail flow, I verify income	- Salary detection from internal accounts (salary-transfer customers) or TPP-aggregated data; documentary fallback (payslips/salary certificate) as refer path - DBR computed with all bureau obligations incl. guarantees; retiree cap switch
As the SME flow, I compute serviceability from transactions	- Categorized business inflows/outflows (TPP or parsed statements); features: avg monthly net inflow, volatility, counterparty concentration, cheque returns - Limit proposal = configurable function of verified net inflows and tenor

8.5 Policy Engine & Policy Console — *[P1, console hardening P2]*

Goal: Credit Risk changes policy in days, under governance, without releases.

User story	Acceptance criteria
As a credit policy manager, I edit policy parameters in a console	- Versioned policy packs per segment; draft → simulate → approve (4-eyes) → schedule publish - Simulation replays a historical application sample and diffs outcomes before publish - Regulatory primitives are locked components – referenced, not editable, by pack authors
As the ISSC/Shari'ah compliance, sector screens are governed	- Exclusion list is a versioned artifact with ISSC approval reference per version; changes follow the agreed delegation matrix

8.6 Scoring — *[P1 v0, P3 custom]*

Goal: risk grades from day one; learning from our own outcomes as fast as volume allows.

User story	Acceptance criteria
As Credit Risk, I launch with a bureau-anchored scorecard	- v0: AECB score bands + application/policy overlays → internal grades; documented per MMS and independently validated before launch
As Credit Risk, I run champion/challenger	- Traffic-split strategies with guardrails; challenger performance dashboards; promotion workflow through the Model Oversight Committee
As a model developer, I build custom scorecards on decision + outcome data	- Decision store + warehouse outcomes joined reproducibly; feature definitions identical in dev and runtime (shared feature store)

8.7 Limits & Pricing Guidance — *[P1]*

Goal: a defensible limit and a compliant pricing band for every approval.

User story	Acceptance criteria
As the retail flow, limits respect all ceilings	- Limit = min(DBR headroom at tenor, 20× salary, product cap); binding constraint recorded in the decision record
As the bank, approvals carry origination-time recourse conditions	- Token conditions are machine-readable: salary + EOSB assignment, takaful enrollment, remittance-linked repayment (cross-border approvals), Kafalah where policy requires - Product layer confirms each condition executed before disbursement; unmet conditions block the token
As the SME flow, facility and drawdown are distinct	- Facility limit at underwriting; drawdown checks (availability, arrears, deterioration flags) < 5s so Tawarruq execution is not blocked

8.8 Decision Management: Refer Workbench & Auditability — [P1]

Goal: fast, consistent, governed human decisions; every decision replayable.

User story	Acceptance criteria
As a credit analyst, I work referred cases from a complete file	- Queues with SLAs (retail same-day, SME < 24h); pre-assembled view: data, features, rules fired, score, recommended action - Overrides require reason codes + second approver; override rates monitored by analyst and rule
As internal audit / Shari'ah audit / CBUAE inspection, I replay any decision	- 100% of decisions reconstructable: input snapshot, feature values, policy & model versions, human actions, linked contract execution events - Retention per regulatory schedule; access controlled and logged

8.9 Monitoring & Governance — [P1 basics, P2 full]

Goal: the platform proves it is working — to us and to the regulator.

User story	Acceptance criteria
As the Head of Credit Risk, I see portfolio decision health	- Dashboards: STP rate, approval/decline mix, refer aging, override rate, decline reason distribution, vintage curves (FPD, 30+ @ 3 MOB) vs appetite
As the Model Oversight Committee, models stay within governance	- Drift monitoring (population stability, feature drift, score-band stability); scheduled bias/fairness stress-tests per the 2026 AI/ML guidance note; validation calendar and findings tracker

8.10 Re-scoring & Limit Management — [P3]

Goal: decisions become a lifecycle, not an event — and distress is acted on at day zero, not DPD 30.

User story	Acceptance criteria
As the bank, I refresh risk on the back book	- Behavioral re-scoring on internal + refreshed bureau data (consent permitting); outputs: limit increase/decrease proposals, early-warning flags to collections
As the bank, I see distress the day it starts	- Day-zero early-warning signals: goal contributions stopping (the earliest sensor), remittance spike, salary interruption before payday, new home-country credit line, balance-depletion velocity - Each signal maps to a same-day agentic action: instalment shifted to salary landing, pre-emptive tenor-matched restructure offer, proactive check-in — signal-to-action time is the tracked metric , because cure rates collapse with latency - Shari'ah constraint holds: restructuring never re-prices the debt
As a product squad, I make pre-approved offers	- Eligible-customer lists with pre-computed limits and expiry; one-tap acceptance re-validates via a lightweight decision call

8.11 Goal-Pursuit & Cross-Border Feature Streams — [P3]

Goal: the strategy's differentiated data — intent telemetry and two-country visibility — enters underwriting as governed features.

User story	Acceptance criteria
As Credit Risk, I use goal-pursuit telemetry as features	- PFM agent streams goal events (contributions, consistency, recovery-after-slippage) into the feature store under a dedicated consented purpose - Features are overlay-only on the explainable core (\$7.7), weighted toward costly-to-fake flows (actual transfers); feature weights never disclosed - Goal-linked financing: a proven saving flow converts one-for-one into an instalment flow — same DBR arithmetic, evidenced affordability
As Credit Risk, I underwrite both countries	- Home-country account data (aggregators / India Account Aggregator where available) and parsed statements (Perfios/Ocrolus-class) feed cross-border income and obligations into affordability - Corridor rollout prioritized by data-rail availability and enforcement reciprocity: India, Philippines, Egypt, Pakistan
As Compliance, the never-list is enforced in code	- Goal/PFM data is technically inaccessible to collections tooling; no third-party disclosure paths exist; access attempts are logged and alerting

9. Roadmap & Phasing

~15–18 months to the Definition of Done. Each phase ends with a **gate** — governance events treated as roadmap items with owners and lead times, because in a Shariah-compliant CBUAE bank they are the critical path as often as engineering is.

Phase	Timeline	Delivers	Gates to exit
P0 — Foundations	M0–M2	Vendor selection (6-week bake-off on our two anchor journeys); decision API contract v1 + canonical application model; AECB institutional integration agreement; TPP partner selection; ISSC engagement opened (delegation matrix proposal); refer-analyst hiring	CBUAE outsourcing non-objection filed; vendor signed with in-UAE deployment; ISSC materiality/delegation framework agreed (this gate unlocks the whole operating model)
P1 — Retail personal finance live	M3–M6	\$8.1–8.3 (consumer), 8.4 (retail), 8.5, 8.6 v0, 8.7, 8.8, 8.9 basics; AECB furnishing pipeline; launch conservative and refer-heavy , tighten with data	ISSC approval of retail policy pack, reason-code language, late-payment handling; MMS-compliant validation of scorecard v0; consumer-protection review (KFS feeds, cooling-off, Arabic/English comms)
P2 — SME working capital live	M6–M10	Commercial + owner-blend bureau; cash-flow serviceability (TPP + statement-upload fallback); SME policy pack + ISSC sector screen; facility/drawdown split; workbench SME queues	ISSC approval of SME pack & screen; SME CPR (Circular 2/2026) compliance review — effective ~H2 2026, so P2 launches into it; credit committee sign-off on SME appetite & guarantee strategy (EDB scheme evaluated)
P3 — Learning loop	M10–M14	Champion/challenger in production; first custom scorecards from own outcomes; re-scoring, day-zero early warning & limit management (\$8.10); goal-pursuit + cross-border feature streams (\$8.11); pre-approved offers; own OFP/TPP authorization business case; risk-based pricing proposal to ISSC	Model Oversight Committee promotion of custom models (intent features as bounded overlays); consent coverage audit for back-book re-scoring; PDPL purpose-consent review of goal telemetry

Phase	Timeline	Delivers	Gates to exit
P4 — Platform proof	M14+	Product onboarding follows the portfolio strategy's learning-speed sequencing (fast-seasoning first): salary-advance (Qard Hassan) pack and covered-card Tawarruq early, SME receivables/invoice finance next, auto finance Murabaha as the < 6-week manifest-only proof; Ijara home refinance deliberately last — a retention product for known customers, never an acquisition anchor; policy simulation at scale	Post-onboarding review: zero core changes required — the platform claim, tested

9.1 Which portfolio first — and keeping the second a first-class citizen

Build the shared layer against retail first — but design its contract against both journeys from day zero. Retail personal finance is the platform's first *consumer*; it is not the *mould* the platform is shaped in.

Why retail leads:

- **Data is API-ready today.** Retail underwrites on the real-time AECB consumer pull plus salary-transfer/WPS signals — both live. SME needs the commercial bureau, owner-blend pulls, bank-statement parsing, and trade-license registries, several still document-based; that integration surface genuinely needs the extra quarters.
- **Volume seasons the platform.** Retail decides at far higher frequency, so the outcome dataset every later capability feeds on — custom scorecards, champion/challenger, re-scoring — accumulates in months, not years. Starting on low-volume, heterogeneous SME would starve the learning loop.
- **The affordability computation is bounded.** Salary-anchored DBR against codified Reg 29/2011 primitives is a tractable first build; SME cash-flow serviceability is judgment-heavy and refer-first, better tackled once the workbench, decision store and monitoring are proven on the easier segment.

The honest counter-argument — and why it doesn't flip the order: SME is where the cross-border thesis and margins are richest, and I inherit the SME Lending owner (\$9.2), so there is real org gravity toward SME-first. But a *platform* is sequenced by learning speed, not strategic value: build where the data flows, prove the abstraction, then bring the high-value segment onto rails that are already load-bearing.

How SME avoids becoming a bolt-on — five mechanisms, not good intentions:

1. **The contract is validated against both anchor journeys before any retail-specific code.** The decision API, canonical application model, and `product manifest` abstraction are specified in P0 against retail *and* SME; retail is simply manifest #1. (The prototype already runs both journeys on one engine — proof the core isn't retail-shaped.)
2. **A "second-product test" is an explicit P1 exit gate.** Retail is not "done" until SME decisions run through the same API in a sandbox — even though SME launches in P2. If the abstraction only fits retail, it fails the gate.
3. **Segment differences are first-class in the model, not retrofits.** The decision record represents *facility + drawdown* (SME) and *single disbursement* (retail) as equals; the limit engine carries both DBR-headroom and inflow-multiple logic from the start; "applicant" is an interface (person *or* entity + owners), never assumed to be a person.
4. **The platform is owned separately from either portfolio** (\$9.2), by someone whose KPI is "products live on the platform," not retail approvals — so platform-agnosticism is explicitly somebody's job.
5. **The SME owner co-authors the shared contract during P0–P1.** The person with the harder, later-shipping segment guards the abstraction while retail is built — the single most effective anti-bolt-on move, and the bridge to the team design below.

9.2 Team & operating model

The target product-owner structure mirrors the architecture: a shared core with two portfolio brains. Steady state (by P3) is three product owners under the Head of Product:

Owner	Owns	Success metric
Shared Credit Platform PO	Decision API, orchestration, consent, data adapters, feature store, decision store, policy console, monitoring – the shared column of \$5.2	Products live on the platform; policy-change lead time; platform reliability; onboarding time for product N
Retail Lending PO	Personal finance, salary advance, auto (future); retail packs, scorecards, affordability, journeys	Retail STP; approval/risk within appetite; time-to-decision
SME Lending PO	Working capital, trade finance; SME packs, cash-flow serviceability, owner-blend, SME workbench	SME STP trajectory; refer-resolution SLA; portfolio quality

The Platform PO is first-among-equals on shared-roadmap arbitration: when retail and SME both want the core changed, the platform owner adjudicates against "does this stay product-agnostic?" – the structural defence against the bolt-on failure mode *and* its opposite, over-fitting the core to whichever portfolio shouts loudest.

Hiring order – reconciling "retail first" with an inherited SME owner:

- **PO (now): me + the inherited SME Lending PO.** We build retail first, but I deliberately place the SME owner at the centre of the *shared contract* definition – co-designing the decision API and feature model so the core is shaped against the harder segment, not just retail. I personally carry retail requirements as player-coach to get P1 moving. This resolves two problems at once: what the SME owner does while retail is built, and how SME stays first-class.
- **Hire 1 (early P1) – Retail Lending PO.** Retail ships first and deserves a dedicated owner; hiring here frees me to lead and keeps the SME owner on platform + SME design rather than being pulled onto retail.
- **Hire 2 (P2, as SME construction starts) – Shared Credit Platform PO.** With two live consumers imminent and a third product on the horizon, the platform needs a permanent, neutral steward. The inherited SME owner now **transitions to own SME Lending end-to-end** – their domain, finally shipping – and the new hire takes the platform. (If the individual's strength and preference lean platform over portfolio, invert it: they keep the platform and hire 2 is the SME PO. Let the person's motivation weigh the call.)
- **Hire 3 (P4, demand-driven, not automatic) – next-wave PO (auto/trade finance)** or a retail split as volume grows. The point of the platform is that product #4 is a *manifest*, not a new team – so this hire is triggered by real load, never by the roadmap by default.

Bringing the inherited report into the new model. Frame it as scope expansion, not disruption: they move from owning one bespoke SME stack to – first – shaping the platform the whole bank will lend on, then (from P2) owning SME as a first-class portfolio on rails they helped design. The risk to manage actively is the perception that retail-first means SME is deprioritised; I counter that concretely with (a) making them the platform-contract co-owner so their segment shapes the core, (b) the P1 second-product test keeping SME visible, and (c) a committed SME launch date (P2), not a vague "later." I'd also give them a genuine voice in the sequencing decision itself – a sequence they helped set is one they defend rather than resent. Career path: position them as the future SME portfolio lead by default, or the Platform PO if that is where their strengths and interest sit.

10. Skeleton Backlog

Epics, mapped to phases and owning workstreams. Stories exist at the level of \$8; this is the planning skeleton.

#	Epic	Workstream	Phase	Note
E01	Decision API contract & orchestration core	Platform eng	P1	Includes approval tokens, idempotency, callbacks
E02	Canonical application & decision data model	Platform eng	PO–P1	The schema products integrate against

#	Epic	Workstream	Phase	Note
E03	Consent registry & evidence store	Platform eng	P1	AECB + AI Tareq scopes
E04	AECB adapter – consumer	Integrations	P1	Real-time pull, cache, no-hit/thin-file handling
E05	AECB furnishing pipeline	Integrations / Compliance	P1	Monthly, quality-checked
E06	AECB adapter – commercial + owner blend	Integrations	P2	Multi-party consent handling
E07	Open-finance ingestion via TPP	Integrations	P1 (retail), P2 (SME)	Lean/Tarabut-class partner; categorization + income verification
E08	Statement upload & parsing fallback	Data eng	P2	SME-critical; also serves AI Tareq coverage gaps
E09	Feature store & shared feature definitions	Data eng	P1	Same definitions at decision-time and model-dev time
E10	Retail affordability (income verification, DBR)	Data eng / Risk	P1	Regulatory primitives embedded
E11	SME cash-flow serviceability engine	Data eng / Risk	P2	Inflow stability, concentration, cheque signals
E12	Policy engine integration & regulatory primitives library	Risk eng	P1	Locked, tested, reusable rule components
E13	Policy console (draft/simulate/approve/publish)	Risk eng	P1-P2	4-eyes, versioning, simulation on historical sample
E14	Shari'ah sector screen (versioned, ISSC-governed)	Risk eng / Shari'ah	P2	AAOIFI SS 21-seeded exclusion list
E15	Scorecard v0 + model registry + MMS documentation	Risk / Data science	P1	Bureau-anchored; independently validated
E16	Champion/challenger framework	Risk / Data science	P3	Traffic split, guardrails, promotion workflow
E17	Limit & pricing engine (retail; SME facility/drawdown)	Risk eng	P1-P2	Banded pricing output
E18	Refer workbench (queues, SLA, overrides, 4-eyes)	Ops tooling	P1	SME queue config in P2
E19	Decision record store & replay	Platform eng	P1	Contract-event linkage for Shari'ah audit
E20	Monitoring & governance dashboards + drift/bias testing	Data / Risk	P1-P2	STP, overrides, vintages; AI/ML guidance-note compliance
E21	Re-scoring & limit management	Risk / Data science	P3	Back-book refresh, early warning, pre-approved offers

#	Epic	Workstream	Phase	Note
E22	Own OFP/TPP authorization track	Product / Legal	P3	Business case then application
E23	Product manifest & onboarding kit ("new product in < 6 weeks")	Platform eng	P4	Config schema, sandbox, playbook
E24	Auto finance onboarding (proof) + Qard Hassan pack	Product squads	P4	Zero core changes is the test
E25	Cross-border data adapters (Credit Passport active path, home-country accounts/statement parsing)	Integrations	P1 (Passport) / P3 (accounts)	The strategy's differentiated data asset
E26	Goal-pursuit telemetry & day-zero early-warning stream	Data eng / Risk	P3	Overlay-only features; never-list enforced in code
E27	Recourse-conditions engine (assignment, takaful, remittance-linked, Kafalah)	Risk eng / Legal	P1 basics / P2 full	Conditions machine-readable on tokens; product layer executes

11. Integration Touch Points

System	Purpose	Mode	Phase	Notes
AECB	Consumer/commercial reports & scores in decisions; monthly furnishing	Real-time API (pull) + batch (furnish)	P1/P2	Subscriber agreement & commercials in P0; telco payment behavior reaches us lawfully through the bureau (never bank-as-operator traffic reading); watch Credit Score 3.0 reason codes for customer comms alignment
Nova Credit / Credit Passport (via AECB)	Home-country bureau history for new-to-UAE applicants – the cross-border approval path	Real-time API, consumer-permissioned per application	P1	India, Philippines, UK live; corridor expansion tracks the strategy's priority list (India, Philippines, Egypt, Pakistan)
Home-country account rails (India Account Aggregator, international open banking, statement parsing)	Cross-border income & obligations into affordability	API via aggregators + document parsing fallback	P3	Perfios/Ocrolus-class parsing; availability varies by corridor – build as adapters, not assumptions
PFM agent platform	Goal-pursuit telemetry (contributions, consistency, plan-vs-actual for SMEs) as governed features; same-day agentic actions on early-warning signals	Event stream, dedicated consented purpose	P3	Overlay-only in scoring (\$7.7); technically walled off from collections
Remittance rails (Mal product)	Remittance-linked repayment sequencing; remittance-spike early-warning signal	Internal events	P2/P3	Instalment settles before the home transfer leaves

System	Purpose	Mode	Phase	Notes
Takaful provider	Credit-shield enrollment (death, disability, job loss) executed as a token condition at origination	API/operational	P2	Priced into the book at origination
AI Tareq / Nebras (via licensed TPP)	Account & transaction data for affordability; future payment initiation	API via TPP (Lean-class)	P1 retail, P2 SME	Own OFP authorization evaluated P3 (ADIB precedent); bank coverage still ramping – never a single point of journey failure
Core banking	Customer master, accounts, facility booking, limit enforcement, disbursement	API/events	P1	Mizan decides limits; core enforces (\$7.6)
Onboarding / KYC / screening stack	Identity (UAE PASS-verified), sanctions/AML outcomes	Event/API consumption	P1	Mizan consumes outcomes; never re-performs KYC
Commodity Murabaha execution platform (DDCAP ETHOS / Eiger-class)	Wakala, commodity purchase/sale, Murabaha contract events	Product-layer API; events linked to decision records	P1	ruya×DDCAP is the live in-market pattern; Mizan needs only event linkage
Statement parsing service	SME bank-statement fallback ingestion	Internal service/vendor	P2	Fraud checks on documents in scope
Trade license registries (NER + emirate DEDs)	License validity, activity codes, age	API via verification vendor (no official API)	P2	Activity codes feed the Shari'ah sector screen
FTA VAT / e-invoicing	Revenue corroboration for larger SME tickets	Documents now; watch e-invoicing APIs as mandate lands 2027	P2+	Lender access rights to e-invoicing data undefined – track, don't depend
EDB Credit Guarantee Scheme	Up to 50% guarantee on qualifying SME facilities (cap AED 5m)	Operational process, partner agreement	P2	Changes loss-given-default in SME policy pack – worth the paperwork
Collections & early-warning	Arrears status into drawdown checks; EWS flags out of re-scoring	Events both ways	P2/P3	Shari'ah constraint: restructuring logic never re-prices
Data warehouse	Decision + outcome data for scorecards, vintages, regulatory reporting	Batch/stream	P1	The learning-loop backbone
CRM / comms	Decision outcomes, adverse-action comms (Arabic + English), KFS data feed	Events/API	P1	Reason-code language ISSC-reviewed

12. Product Metrics

Metric	Target	Why it matters
Straight-through processing	Retail ≥ 80% by M9 · SME ≥ 40% by M12, ≥ 60% by M18	The automation promise, segment-honest

Metric	Target	Why it matters
Time to decision	Retail < 5 min p90 · SME refer resolution < 24h, < 1h by M14	Customer promise; competitive parity with digital natives
Policy change lead time	< 1 week idea→production (within delegation matrix)	The self-service promise — measures the org, not just the tool
Decision replay coverage	100% from day one	Non-negotiable audit posture
Vintage risk (FPD, 30+ @ 3 MOB)	Within CRO-set appetite corridors per segment	Proves automation isn't buying growth with risk
Override rate	< 10% of refers overridden without pattern; overrides feed ≥ 1 policy change/quarter	Human-judgment loop is working
Cost per decision	Baseline in P1, -30% by P3	Bureau + data + platform amortization
Third-product onboarding time	< 6 weeks (P4 test)	The platform claim, measured
Cross-border data attach rate on thin files	≥ 60% of no-hit applicants underwritten via Credit Passport or home-account data by M12	The strategy's approval segment, measured — thin files become approvals, not exceptions
Signal-to-action time (early warning)	Same-day (D+0) from signal to first agentic action, from P3	The dossier's core claim: the moat is not the signal, it is signal-to-action latency — cure rates collapse with every week
Goal-telemetry coverage	≥ 40% of active borrowers with consented goal streams by M18	Feeds both underwriting overlays and the earliest distress sensor

13. Risks & Mitigations

Risk	Severity	Mitigation
Thin-file / new-to-UAE applicants dominate retail demand (expat market)	High	Explicit thin-file strategy from P1: AECB Credit Passport (consented home-country history), conservative starter limits, salary-transfer-anchored offers; AECB Score 3.0 alt-data scores widen coverage through 2026
AI Tareq coverage ramps slower than planned	Medium	Never a hard dependency: TPP + internal data + statement upload; roadmap assumes documents for SME at P2 launch
ISSC approval cadence throttles policy agility	High	The PO delegation-matrix gate; ISSC engaged on architecture (not just products) from month 1; pre-agreed artifact formats for screen/policy versions
Vendor lock-in / vendor underdelivers	Medium	Own the API contract, feature definitions, decision store; bake-off on real journeys before signing; exit provisions + data egress tested annually
SME data sparsity → weak early SME models	Medium	Refer-first launch posture; human decisions structured (reason-coded) so they become training data; EDB guarantees de-risk the learning period
Model governance debt (MMS, AI/ML guidance) discovered late	Medium	Governance artifacts are epic-level scope (E15, E20) with the same priority as features; Model Oversight Committee stood up in PO

Risk	Severity	Mitigation
Regulatory drift (SME CPR effective ~H2 2026; new Central Bank Law transition to Sep 2026)	Medium	Compliance review is a named gate in every phase; regulatory primitives centralized so a rule change is one edit, not N product edits
Goodhart risk on intent features — once goal pursuit is a known credit input, it gets simulated	Medium	Weight only costly-to-fake flows (actual transfers, actual supplier payments); never disclose feature weights; monitor feature drift for gaming signatures
Honesty-punishment data poisoning — if declared departures or goal changes are penalized, disclosure stops and the intent stream degrades	High	Encoded rule: score inconsistency-without-cause, not goal changes; declared departure triggers a tenor-matched product offer, never a price penalty; same for documented life shocks
Trust/conduct catastrophe — one misuse of service data as collections pressure ends the flywheel and invites regulatory action	Low likelihood / severe	The never-list (no family disclosure, no contact-graph leverage, no surveillance telemetry) enforced as technical access walls + PDPL purpose limitation; ISSC oversight as independent kill-switch

14. Dependencies

Team / Party	What they own or provide
Credit Risk (CRO org)	Risk appetite, policy pack content, scorecard sign-off, refer-queue staffing; co-owner of the policy console
ISSC & Shari'ah Compliance	Delegation matrix, product/policy fatwas, sector-screen approval, audit of contract-event sequencing
Compliance & Legal	CBUAE outsourcing non-objection, AECB agreement, consumer-protection & SME CPR reviews, TPP contract
Core Banking team	Facility booking, limit enforcement, disbursement APIs
Onboarding/KYC team	Identity & screening outcomes as consumable events
Data Platform team	Warehouse, outcome pipelines, feature-store infrastructure
Retail & SME product squads	Journey UX, contract execution integration, KFS/comms surfaces
Finance / Treasury	Funding cost inputs to pricing bands; EDB guarantee commercials

15. Open Questions

#	Question	Context
1	What is Mal Bank's exact license type?	A Specialised Bank (Low-Risk) license caps total assets (AED 25bn), restricts business to AED and residents, and limits wholesale/SME lending to 50% of the book — this directly sizes the SME portfolio and could reorder P2 vs P3
2	What decisioning capability exists today, and what must be migrated vs sunset?	The brief says "largely bespoke" — a migration/parallel-run plan for in-flight portfolios needs discovery

#	Question	Context
3	AECB commercial terms and API SLAs for a digital-first volume profile?	Per-pull economics shape caching strategy and pre-qualification design (soft-check availability)
4	Which TPP, and what is real AI Tareq bank coverage today?	Determines how much SME affordability is API-fed vs document-fed at P2 launch
5	Will the ISSC accept a materiality-based delegation matrix for policy changes?	The single biggest determinant of the "policy in days" goal; needs the first ISSC conversation, not the last
6	Risk-based pricing ambition — does Mal Bank want it at all?	Banded pricing may be a durable brand position for a community-oriented Islamic bank, not just a launch constraint
7	Where does salary advance (Qard Hassan) sit commercially?	It cannot price for risk (SS 19); as an engagement/retention product its decisioning is trivial — but its funding cost and loss appetite need a business owner
8	Collections & restructuring interface	Shari'ah constraints (no re-pricing, donation-based late payments) mean collections strategy must be co-designed with the decisioning layer's data — whose roadmap?
9	Corridor sequencing and rail readiness	India/Philippines/Egypt/Pakistan is the strategy's priority order — but Credit Passport coverage, Account Aggregator access mechanics for a UAE bank, and enforcement reciprocity (2020 India notification) need per-corridor validation before committing P3 scope
10	ISSC position on goal-linked financing and intent features	Converting a saving flow into an instalment flow, and using behavioral telemetry in credit decisions, both need early ISSC socialization — fairness remit applies; this belongs in the P0 delegation-matrix conversation
11	PDPL consent architecture for service-data-as-credit-input	Goal telemetry needs its own purpose-specific consent, separable from PFM service consent and revocable without losing the service — legal design needed before P3 build

16. What I'd Go Deeper On With More Time

- 1. Vendor bake-off specification** — turn the two anchor journeys into an executable evaluation script (data payloads, policy pack, expected outcomes) and run it against 3 shortlisted platforms.
- 2. SME limit model design** — the inflow-multiple heuristic needs segmentation by sector/seasonality and back-testing against any historical SME book Mal Bank has.
- 3. Thin-file strategy economics** — quantify the approvable-but-invisible population (new-to-UAE, no-borrowing-history) and the revenue case for Credit Passport + starter products.
- 4. Consent UX research** — AI Tareq's centralized consent adds an authentication hop; measure drop-off and design recovery paths.
- 5. Migration & parallel-run plan** — shadow-decisioning against the incumbent logic before cutover, with divergence analysis as the acceptance test.
- 6. Collections co-design** — early-warning signals from re-scoring into a Shari'ah-compliant treatment framework.
- 7. Pricing governance** — a full proposal to the ISSC on band structures, EIBOR-benchmarked markup mechanics, and fairness monitoring.

Appendix A — Research Base (selected sources)

Regulatory (CBUAE): Open Finance Regulation — Circular 3/2025 (replacing 7/2023), CBUAE Rulebook; Nebras/AI Tareq operator materials; Consumer Protection Regulation 8/2020 + Standards (cooling-off, responsible financing Art. 7); Regulation 29/2011 (20× salary, 48-month tenor, DBR 50%/30%, auto 80%/60); SME Customer Protection Regulation — Circular 2/2026; Model Management Standards — Notice 5052/2022; Guidance Note on Responsible AI/ML — 11 Feb 2026; Outsourcing Regulation & Standards (2021); Specialised Banks (Low-Risk) Regulation; Federal Decree-Law 6/2025 (new Central Bank Law).

Credit bureau: AECB products (consumer/commercial reports & scores 300–900, ESR, ChequeScore); Credit Passport with Nova Credit (2023); Credit Score 3.0 (announced Nov 2025 for H1 2026); real-time API precedent (Policybazaar, 2021); Federal Law 6/2010 on Credit Information (consent); monthly furnishing obligation (Rulebook credit-report articles); 119 data providers incl. telcos/utilities/Ministry of Justice.

Open finance market: ADIB — first Islamic bank live under AI Tareq (Jan 2026) and first bank OFP/TPP license (Apr 2026); FAB & CBD go-lives with Pay10 (Jan 2026); Lean Technologies (CBUAE-licensed TPP, live-proving graduate); Tarabut (in-principle approval Aug 2025); Lean×Ziina Pay-by-Bank (Jun 2026).

Shari'ah: AAOIFI SS 8 (Murabaha), SS 30 (Tawarruq), SS 19 (Qard fees), SS 3 (late-payment donation), SS 59 (sale of debt), SS 21 (screening taxonomy); HSA Resolution 18/3/2018 (AAOIFI mandatory); CBUAE Shari'ah Governance Standard (ISSC scope incl. operational process flows); Dubai Court of Cassation 595/2025 and Cassation 9/2025 (late-payment interest void; "interest"-labelled clauses void); UAE Commercial Transactions Law 50/2022 Arts. 468–497.

Market/vendors: ruya × DDCAP ETHOS commodity Murabaha automation (Jul 2025); Eiger Trading REST-API Tawarruq execution; Emirates Islamic 3-hour Murabaha offer window; ADIB Instant Finance category-uniform pricing; Wio SME lending (Network International marketplace, Geidea); Biz2X × Network International / Deem Finance; FICO × Network International in-region hosting; Experian Middle East (Dubai); Provenir MEA expansion; Taktile (no public MEA presence); EDB Credit Guarantee Scheme (50%, AED 5m cap); UAE e-invoicing Ministerial Decisions 243/244 of 2025 (mandatory from 2027); Cabinet Resolution 22/2016 (SME definition).

Strategy alignment: this PRD implements the decisioning layer for the credit portfolio strategy set out in the companion [Cross-Border Underwriting Dossier](#) (working dossier v3, 21 Jul 2026) — cross-border thesis, future-intent underwriting, learning-speed sequencing, day-zero early warning, consent-first recourse, and the telco-via-bureau position.

Assessment note: facts above were verified via public sources in July 2026; items flagged in-text (e.g., exact SME CPR effective date, AECB commercial terms, Credit Score 3.0 go-live status) require primary-source confirmation during discovery.